

1. SAFETY

Safety first!

1. Personal grooming:

- Never wear loose clothes when operating machines; they can be caught in revolving parts of the machine
- Remove wristwatches, rings, bracelets, necklaces, or other jewelry,
- Never wear gloves when operating machines; these can be caught in rotating parts of machines or the workpiece.
- Long hair must be covered or protected by a hair net or approved protective shop cap.
- Canvas shoes or open-toe sandals must never be worn in a machine shop.

2. Work place environment

- Always use a brush or vacuum to remove any chips; never use cloth or hand.
- Clean oily surfaces and tools with a dry cloth.
- Tools and materials should not be stored on the machine. A serious accident may occur if they fall.
- To prevent slipping or falling accidents, always clean oil, grease, or other liquids from the floor around the machine.
- Metal chips on the floor should be swept up frequently. Chips can become embedded in the soles or heels of shoes and can cause dangerous walking conditions and/or slippage.
- Keep the floor areas close to machines free of tools or materials.
- After cutting off stock to the required length, return the bar stock to its proper storage rack,
- Never use compressed air to blow chips away from a machine.

3. Safe work practice

When actually operating tools and machines, the following practices should be observed:

- Do not attempt to operate any machine or power tool equipment before understanding how it works and how to stop it quickly if something unexpected happens.
- Make sure all guards and safety devices are in place and operational before operating any equipment

- Always disconnect the electric power and lock it off at the switch box when making repairs or adjustments to any equipment.
- Check to be sure that the work piece and cutting tools are properly secured before operating the machine.
- Keep hands away from moving parts. never attempt to "feel" the surface of the revolving work, or to stop the machine by hand.
- Never measure, clean, or make any adjustment unless the machine has come to a complete stop.
- Rags should never be used near moving parts as both the hand and rag can be drawn in to the machine.
- Only one person should operate a machine at any one time.
- Every injury, regardless of how minor, should get first-aid attention immediately. Even the smallest cut should be treated to prevent infection.
- Always remove all sharp burrs from work pieces with a file before handling.
- Get help before attempting to lift anything too heavy or odd-shaped and follow safe lifting practice.
- Always make sure that the work is clamped securely before attempting a machining operations.
- Never start a machine until you are certain that the work piece will clear the cutting tool and machine parts.
- Use a proper fitting wrench for each job an over size wrench can lead to worn corners on the nut. Worn nuts should always be replaced.
- Pulling a wrench is much safer than pushing it. Pushing on wrenches can be dangerous, especially if the hands or wrench are greasy.
- During grinding, use coolant whenever possible to control the grinding dust.

2. CONVENTIONAL MACHINING

Conventional machining, in which a sharp cutting tool is used to mechanically cut the material to achieve the desired geometry.

Machining is a manufacturing process in which a sharp cutting tool is used to cut away material to leave the desired part shape. The predominant cutting action in machining involves shear deformation of the work material to form a chip; as the chip is removed, a new surface is exposed. Machining is most frequently applied to shape metals. The process is illustrated in the diagram of Figure 1.

Even if there are many type of machines, here we will discuss on the most common ones.

2.1. LATHE

Definition:-Lathe is a machine, which removes the metal from a piece of work to obtain the required shape and size. Lathe is the oldest machine tool.

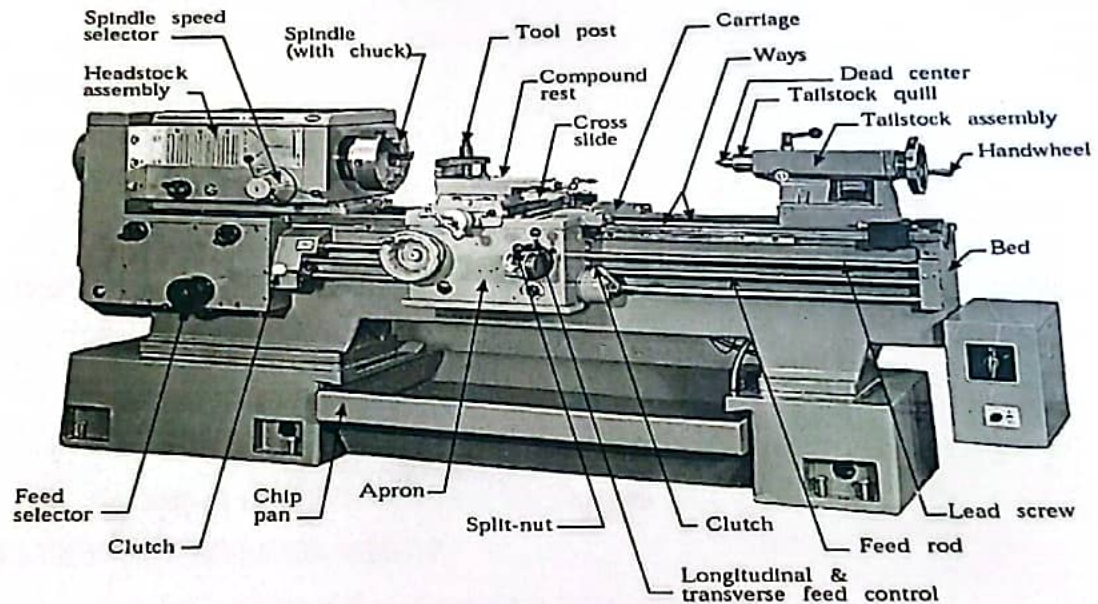


Fig (1) Engine Lathe

Lathe specifications

- Distance between centers
- Swing over the bed

- Swing over the cross slide
- Horse power of the motor
- Number of speeds range
- Number of feeds

Lathe Components

❑ LATHE BED

- The bed is the base of the lathe and supports all the major components of lathe.
- Lathe bed material made of grey cast iron , to resist deflection and absorb vibrations during cutting

❑ CARRIAGE:

- slides along the ways and consists of the *cross-slide, tool post, apron*
- Carriage Feed
 - Longitudinal Feed or “Turning” - The tool is fed *along* the work.
 - Cross Feed or “Facing” – The tool is fed *across* the work.

❑ HEADSTOCK

- Holds the jaws for the work piece,
- supplies power to the jaws and
- provides various drive Speeds

❑ TAIL STOCK:

- supports the other end of the work piece
- It's like a stationary drill press
- It is centered with your work piece
- For drilling use a drill chuck that fits your bits

❑ FEED ROD AND LEAD SCREW

- Feed rod is powered by a set of gears from the headstock

Cutting Tools

There are basically two types of cutting tools:

- Single point (e.g. turning tools). (fig .2)

- Multiple point (e.g. milling tools).

Fig (2) shows single point cutting tool

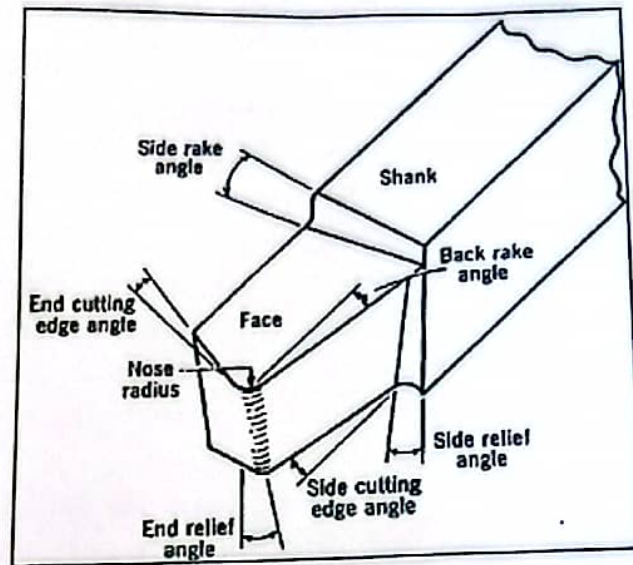


Fig (2) Single Point Cutting Tool

Lathe operations

- a) **Parallel turning:** the tool is fed parallel to the axis of the rotating work piece to produce cylindrical component or feature.

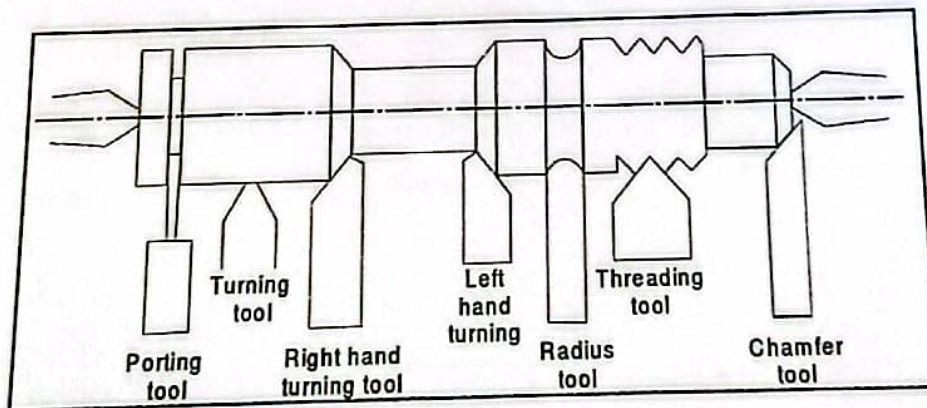
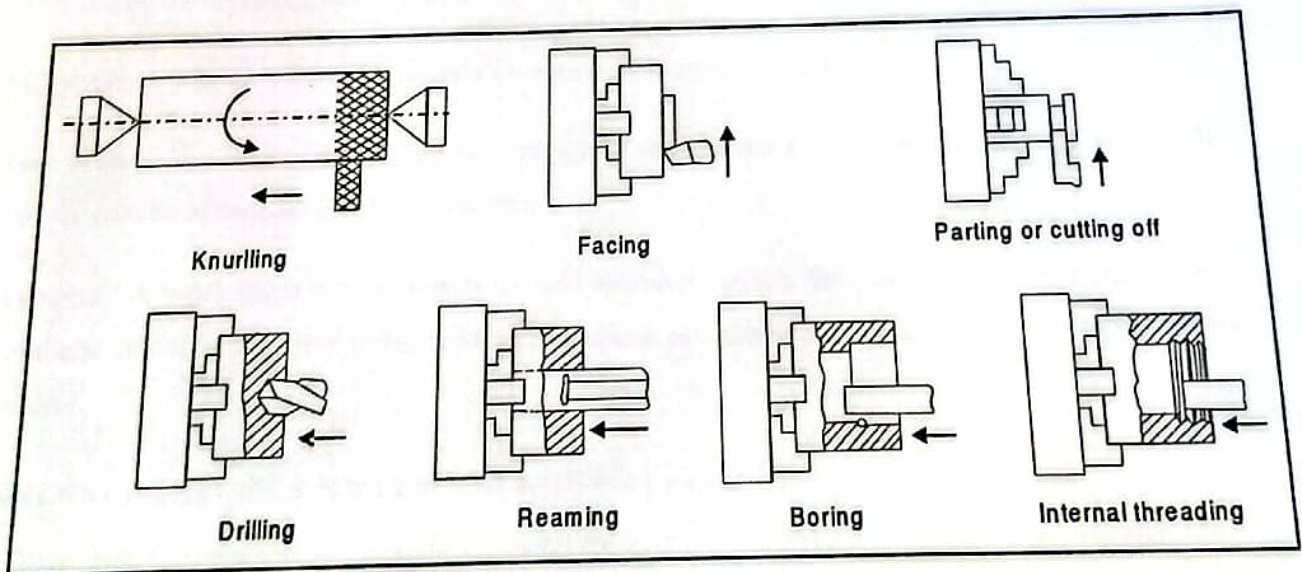


Fig.3. Different types of lathe operations

- b) **Facing:** The tool is fed radially into the rotating work on one end to create a flat surface on the end.

- c) **Taper turning:** Instead of feeding the tool parallel to the axis of rotation of the work, the tool is fed at an angle, thus creating a tapered cylinder or conical shape.
- d) **Contour turning:** Instead of feeding the tool along a straight line parallel to the axis of rotation as in turning, the tool follows a contour that is other than straight, thus creating a contoured form in the turned part.
- e) **Form turning:** In this operation, sometimes called forming, the tool has a shape that is imparted to the work by plunging the tool radially into the work.
- f) **Chamfering:** The cutting edge of the tool is used to cut an angle on the corner of the cylinder, forming what is called a "chamfer."
- g) **Cutoff:** The tool is fed radially into the rotating work at some location along its length to cut off the end of the part. This operation is sometimes referred to as parting.
- h) **Threading:** A pointed tool is fed linearly across the outside surface of the rotating workpart in a direction parallel to the axis of rotation at a large effective feed rate, thus creating threads in the cylinder.
- i) **Boring:** A single-point tool is fed linearly, parallel to the axis of rotation, on the inside diameter of an existing hole in the part.
- j) **Drilling:** Drilling can be performed on a lathe by feeding the drill into the rotating work along its axis. Reaming can be performed in a similar way.
- k) **Knurling:** This is not a machining operation because it does not involve cutting of material. Instead, it is a metal forming operation used to produce a regular crosshatched pattern in the work surface.



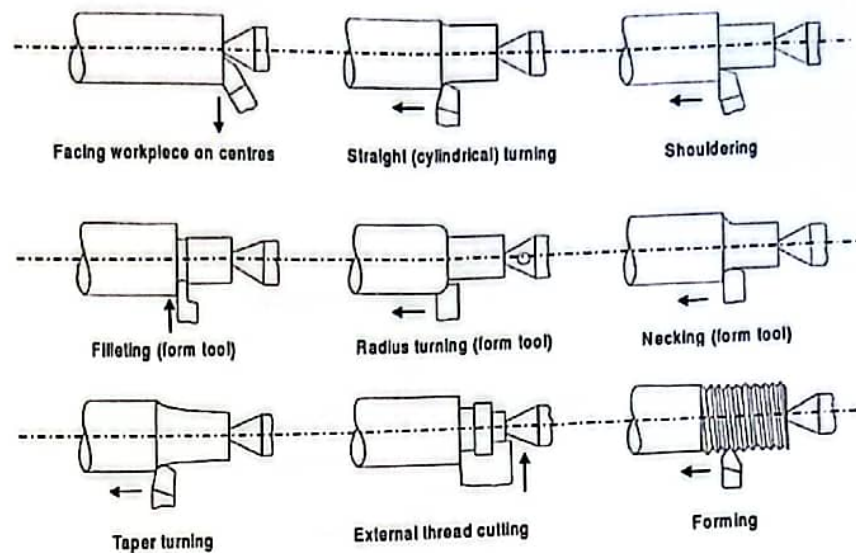


Fig (4) Machining operations other than turning that are performed on a lathe:

Work holding Devices for Lathes:

Work pieces can be held by various methods.

Three Jaw chuck: It usually has three jaws, the jaws are moved simultaneously within the chuck.

Four Jaw chuck: This is independent chuck generally has four jaws , which are adjusted individually on the chuck face by means of adjusting screws.

Magnetic chuck: Thin jobs can be held by means of magnetic chucks.

Face plates: The face plate is used for irregularly shaped work pieces that cannot be successfully held by chucks or mounted between centers.

Mandrels: A work piece which cannot be held between centers because its axis has been drilled or bored and which is not suitable for holding in a chuck or against a faceplate is usually machined on a mandrel.

Collet chuck: Collet chuck is used to hold small work pieces.

CUTTING CONDITIONS IN TURNING

The rotational speed in turning is related to the desired cutting speed at the surface of the cylindrical workpiece by the equation

$$N = \frac{v}{\pi D_o}$$

where N = rotational speed, rev/min; v = cutting speed, m/min (ft/min); and D_o = original diameter of the part, m (ft).

The turning operation reduces the diameter of the work from its original diameter D_o to a final diameter D_f , as determined by the depth of cut d :

$$D_f = D_o - 2d$$

The feed in turning is generally expressed in mm/rev (in/rev). This feed can be converted to a linear travel rate in mm/min (in/min) by the formula

$$f_r = Nf$$

where f_r = feed rate, mm/min (in/min); and f = feed, mm/rev (in/rev). The time to machine from one end of a cylindrical workpart to the other is given by

$$T_m = \frac{L}{f_r}$$

where T_m = machining time, min; and L = length of the cylindrical workpart, mm (in). A more direct computation of the machining time is provided by the following equation:

$$T_m = \frac{\pi D_o L}{f v}$$

Where;

D_o = work diameter, mm; L = workpart length, mm; f = feed, mm/rev (in/rev); and v = cutting speed, mm/min (in/min). As a practical matter, a small distance is usually added to the work part length at the beginning and end of the piece to allow for approach and overtravel of the tool. Thus, the duration of the feed motion past the work will be longer than T_m .

Taper Turning

$$\tan \alpha = (D_1 - D_2) / 2L \quad \text{where } \alpha = \text{angle of taper}$$

D_1 = major diameter in mm

D_2 = minor diameter in mm

L = Length of taper in mm

The conicity K of the taper is defined as $K = (D_1 - D_2) / L$

Lathe simple problems

Problem -1

A mild steel rod having 50 mm diameter and 500 mm length is to be turned on a lathe. Determine the machining time to reduce the rod to 45 mm in one pass when cutting speed is 30 m/min and a feed of 0.7 mm/rev is used.

Solution

Given data: $D = 50$ mm, $L_j = 500$ mm

$v = 30$ m/min, $f = 0.7$ mm/rev

Substituting the values of v and D in $V = \pi DN / 1000$ M/min

Required spindle speed as: $N = 191$ rpm

Problem -2

Determine the angle at which the compound rest would be swiveled for cutting a taper on a work piece having a length of 150 mm and outside diameter 80 mm. The smallest diameter on the tapered end of the rod should be 50 mm and the required length of the tapered portion is 80 mm.

Solution

Given data: $D_1 = 80$ mm, $D_2 = 50$ mm, $L_j = 80$ mm (with usual notations) $\tan \alpha = (80 - 50) / 2 \times 80$
or $\alpha = 10.620$

The compound rest should be swiveled at 10.62°